Foundation Mathematics 1017SCG
Week 2 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a)

$$\log_4(16) = 2$$
$$(\text{as } 4^2 = 16)$$

(b)

$$\log_7(7) = 1$$
$$(\text{as } 7^1 = 7)$$

(c)

$$\log_5(125) = 3$$
$$(\text{as } 5^3 = 125)$$

(d)

$$\log_9(1) = 0$$
$$(\text{as } 9^0 = 1)$$

(e)

$$\begin{aligned}\log_3\left(\frac{1}{3}\right) &= \log_3(1) - \log_3(3)\\ &= 0 - 1\\ &= -1\end{aligned}$$

(f)

$$\begin{aligned}\log_5\left(\frac{1}{25}\right) &= \log_5(1) - \log_5(25)\\ &= 0 - 2\\ &= -2\end{aligned}$$

(g)

$$\begin{aligned}\log_2\left(\frac{1}{8}\right) &= \log_2(1) - \log_2(8)\\ &= 0 - 3\\ &= -3\end{aligned}$$

2. (a)

$$\begin{aligned}\log_{12}(6) + \log_{12}(2) &= \log_{12}(6 \times 2)\\ &= \log_{12}(12)\\ &= 1\end{aligned}$$

(b)

$$\begin{aligned}\log_8(2) + \log_8(4) &= \log_8(2 \times 4)\\ &= \log_8(8)\\ &= 1\end{aligned}$$

(c)

$$\begin{aligned}\log_6(3) + \log_6(12) &= \log_6(3 \times 12)\\ &= \log_6(36)\\ &= 2\end{aligned}$$

(d)

$$\begin{aligned}\log_4\left(\frac{1}{2}\right) + \log_4(32) &= \log_4\left(\frac{1}{2} \times 32\right)\\ &= \log_4(16)\\ &= 2\end{aligned}$$

(e)

$$\begin{aligned}\log_5(8) + \log_5\left(\frac{1}{8}\right) &= \log_5\left(8 \times \frac{1}{8}\right)\\ &= \log_5(1)\\ &= 0\end{aligned}$$

(f)

$$\begin{aligned}\log_3(54) - \log_3(6) &= \log_3\left(\frac{54}{6}\right)\\ &= \log_3(9)\\ &= 2\end{aligned}$$

(g)

$$\begin{aligned}\log_5(250) - \log_5(2) &= \log_5\left(\frac{250}{2}\right)\\ &= \log_5(125)\\ &= 3\end{aligned}$$

(h)

$$\begin{aligned}\log_9(90) - \log_9(5) - \log_9(2) &= \log_9\left(\frac{90}{5}\right) - \log_9(2)\\ &= \log_9(18) - \log_9(2)\\ &= \log_9\left(\frac{18}{2}\right)\\ &= \log_9(9)\\ &= 1\end{aligned}$$

(i)

$$\begin{aligned}\frac{\log_6(x^5)}{\log_6(x)} &= \frac{5 \times \log_6(x)}{\log_6(x)} \\ &= 5\end{aligned}$$

(j)

$$\begin{aligned}\frac{\log_4(\sqrt{x})}{\log_4(x)} &= \frac{\log_4(x^{\frac{1}{2}})}{\log_4(x)} \\ &= \frac{\frac{1}{2} \times \log_4(x)}{\log_4(x)} \\ &= \frac{1}{2}\end{aligned}$$

3. Most scientific calculators can only calculate logarithms that are either base 10 or base e. To convert to one of these bases, use the change of base rule. In these solutions, the change of base rule has been used to change to base 10. However, you could have also chosen to use base e.

(a)

$$\log_{10}(123) = 2.0899$$

(b)

$$\ln(40) = 3.6889$$

(c)

$$\begin{aligned}\log_5(60) &= \frac{\log_{10}(60)}{\log_{10}(5)} \\ &= 2.5440\end{aligned}$$

(d)

$$\begin{aligned}\log_7(25) &= \frac{\log_{10}(25)}{\log_{10}(7)} \\ &= 1.6542\end{aligned}$$

(e)

$$\begin{aligned}\log_2(18) &= \frac{\log_{10}(18)}{\log_{10}(2)} \\ &= 4.1699\end{aligned}$$

(f)

$$\begin{aligned}\log_9(2) &= \frac{\log_{10}(2)}{\log_{10}(9)} \\ &= 0.3155\end{aligned}$$

4. In this question, we can use Pythagoras' Theorem to find the length of the unknown side of the right-angled triangle.

(a)

$$\begin{aligned} 5^2 + 8^2 &= x^2 \\ 25 + 64 &= x^2 \\ 89 &= x^2 \\ x &= \sqrt{89} \\ x &= 9.43\text{m} \end{aligned}$$

(b)

$$\begin{aligned} 3^2 + 7^2 &= x^2 \\ 9 + 49 &= x^2 \\ 58 &= x^2 \\ x &= \sqrt{58} \\ x &= 7.62\text{m} \end{aligned}$$

(c)

$$\begin{aligned} x^2 + 14^2 &= 20^2 \\ x^2 + 196 &= 400 \\ x^2 + 196 - 196 &= 400 - 196 \\ x^2 &= 204 \\ x &= \sqrt{204} \\ x &= 14.28\text{cm} \end{aligned}$$

(d)

$$\begin{aligned} x^2 + 40^2 &= 55^2 \\ x^2 + 1600 &= 3025 \\ x^2 + 1600 - 1600 &= 3025 - 1600 \\ x^2 &= 1425 \\ x &= \sqrt{1425} \\ x &= 37.75\text{mm} \end{aligned}$$

5. (a)

$$\begin{aligned} \sin(40^\circ) &= \frac{x}{9} \\ \sin(40^\circ) \times 9 &= \frac{x}{9} \times 9 \\ \sin(40^\circ) \times 9 &= x \\ x &= 5.79\text{m} \end{aligned}$$

(b)

$$\cos(35°) = \frac{x}{15}$$
$$\cos(35°) \times 15 = \frac{x}{15} \times 15$$
$$\cos(35°) \times 15 = x$$
$$x = 12.29\text{cm}$$

(c)

$$\tan(28°) = \frac{x}{38}$$
$$\tan(28°) \times 38 = \frac{x}{38} \times 38$$
$$\tan(28°) \times 38 = x$$
$$x = 20.20\text{mm}$$

(d)

$$\cos(41°) = \frac{5}{x}$$
$$\cos(41°) \times x = \frac{5}{x} \times x$$
$$\cos(41°) \times x = 5$$
$$\frac{\cos(41°) \times x}{\cos(41°)} = \frac{5}{\cos(41°)}$$
$$x = 6.63\text{km}$$

6. (a)

$$\sin(\theta) = \frac{4}{6}$$
$$\theta = \sin^{-1}\left(\frac{4}{6}\right)$$
$$\theta = 41.81°$$

(b)

$$\tan(\theta) = \frac{18}{25}$$
$$\theta = \tan^{-1}\left(\frac{18}{25}\right)$$
$$\theta = 35.75°$$

7. (a) We will start by finding the length of side b. Alternatively, this can be written as the length of AC.

$$\tan(50°) = \frac{18}{b}$$
$$\tan(50°) \times b = \frac{18}{b} \times b$$
$$\tan(50°) \times b = 18$$
$$\frac{\tan(50°) \times b}{\tan(50°)} = \frac{18}{\tan(50°)}$$
$$b = 15.10\text{cm}$$

We can now find the length of side c using the sin ratio. Alternatively, this could also be done using Pythagoras' Theorem.

$$\begin{aligned}
\sin(50^\circ) &= \frac{18}{c} \\
\sin(50^\circ) \times c &= \frac{18}{c} \times c \\
\sin(50^\circ) \times c &= 18 \\
\frac{\sin(50^\circ) \times c}{\sin(50^\circ)} &= \frac{18}{\sin(50^\circ)} \\
c &= 23.50\text{cm}
\end{aligned}$$

Finally, we can find the size of angle B using the knowledge that the sum of all angles in a triangle is 180°.

$$\begin{aligned}
B &= 180^\circ - 90^\circ - 50^\circ \\
&= 40^\circ
\end{aligned}$$

(b) We will start by using Pythagoras to find length of side p.

$$\begin{aligned}
p^2 + 19^2 &= 28^2 \\
p^2 + 361 &= 784 \\
p^2 + 361 - 361 &= 784 - 361 \\
p^2 &= 423 \\
p &= \sqrt{423} \\
p &= 20.57
\end{aligned}$$

Next, we will find the size of angle P.

$$\begin{aligned}
\cos(P) &= \frac{19}{28} \\
P &= \cos^{-1}\left(\frac{19}{28}\right) \\
P &= 47.27^\circ
\end{aligned}$$

Finally, the size of angle R can be found.

$$\begin{aligned}
R &= 180^\circ - 90^\circ - 47.27^\circ \\
&= 42.73^\circ
\end{aligned}$$

(c) It is important to note that the length of side s is the same as the length of side u. This is indicated by the small perpendicular line through side s and side u of the triangle. A triangle where two of its sides have the same length is called an isosceles triangle. As it is an isosceles triangle, the size of angle S is the same as the size of angle U. Therefore $S = 45^\circ$ and $U = 45^\circ$ (as $45^\circ + 45^\circ + 90^\circ = 180^\circ$).

We will now find the length of side s (which is the same as the length of side u).

$$s^2 + u^2 = 70^2$$

However, as the length of side s is the same as the length of side u, we have

$$
\begin{aligned}
s^2 + s^2 &= 70^2 \\
2s^2 &= 4900 \\
\frac{2s^2}{2} &= \frac{4900}{2} \\
s^2 &= 2450 \\
s &= \sqrt{2450} \\
s &= 49.50\text{mm}
\end{aligned}
$$

Therefore $s = 49.50$mm and $u = 49.50$mm.

Above, we used the fact that it was an isosceles triangle to determine that the size of angles S and U is 45°. However, you could also calculate the size of angle S using the working below.

$$
\begin{aligned}
\sin(S) &= \frac{49.50}{70} \\
S &= \sin^{-1}\left(\frac{49.50}{70}\right) \\
S &= 45.00^\circ
\end{aligned}
$$

(d) It is important to recognise that this triangle is NOT a right-angled triangle.
We will start by finding the size of angle A.

$$A = 180 - 55 - 47 = 78^\circ$$

Now we will calculate the length of side c using the sine rule.

$$
\begin{aligned}
\frac{c}{\sin(47^\circ)} &= \frac{20}{\sin(55^\circ)} \\
\frac{c}{\sin(47^\circ)} \times \sin(47^\circ) &= \frac{20}{\sin(55^\circ)} \times \sin(47^\circ) \\
c &= 17.86\text{cm}
\end{aligned}
$$

Finally, we will calculate the length of side a using the sine rule.

$$
\begin{aligned}
\frac{a}{\sin(78^\circ)} &= \frac{20}{\sin(55^\circ)} \\
\frac{a}{\sin(78^\circ)} \times \sin(78^\circ) &= \frac{20}{\sin(55^\circ)} \times \sin(78^\circ) \\
a &= 23.88\text{cm}
\end{aligned}
$$

(e) We will start by calculating the size of angle Y using the sine rule.

$$
\begin{aligned}
\frac{\sin(Y)}{31} &= \frac{\sin(140^\circ)}{180} \\
\frac{\sin(Y)}{31} \times 31 &= \frac{\sin(140^\circ)}{180} \times 31 \\
\sin(Y) &= \frac{\sin(140^\circ)}{180} \times 31 \\
Y &= \sin^{-1}\left[\frac{\sin(140^\circ)}{180} \times 31\right] \\
Y &= 6.36^\circ
\end{aligned}
$$

Therefore, the size of angle Z is given by

$$
\begin{aligned}
Z &= 180^\circ - 140^\circ - 6.36^\circ \\
&= 33.64^\circ
\end{aligned}
$$

Finally, we can calculate the length of side z

$$
\begin{aligned}
\frac{z}{\sin(33.64^\circ)} &= \frac{180}{\sin(140^\circ)} \\
\frac{z}{\sin(33.64^\circ)} \times \sin(33.64^\circ) &= \frac{180}{\sin(140^\circ)} \times \sin(33.64^\circ) \\
z &= \frac{180}{\sin(140^\circ)} \times \sin(33.64^\circ) \\
z &= 155.13\text{mm}
\end{aligned}
$$

Due to rounding errors, your final answer for side z may be slightly different to what is shown above.

(f) We will start by finding the length of side c using the cosine rule.

$$c^2 = 8^2 + 15^2 - 2 \times 8 \times 15 \times \cos(66^\circ)$$

Using our scientific calculator, we can evaluate the right-hand side of the above equation all in one step.

$$
\begin{aligned}
c^2 &= 191.38 \\
c &= \sqrt{191.38} \\
c &= 13.83\text{cm}
\end{aligned}
$$

We will now find the size of angle A.

$$
\begin{aligned}
\frac{\sin(A)}{15} &= \frac{\sin(66^\circ)}{13.83} \\
\frac{\sin(A)}{15} \times 15 &= \frac{\sin(66^\circ)}{13.83} \times 15 \\
\sin(A) &= \frac{\sin(66^\circ)}{13.83} \times 15 \\
A &= \sin^{-1}\left[\frac{\sin(66^\circ)}{13.83} \times 15\right] \\
A &= 82.23^\circ
\end{aligned}
$$

Finally, we can find the size of angle B.

$$
\begin{aligned}
B &= 180^\circ - 66^\circ - 82.23^\circ \\
&= 31.77^\circ
\end{aligned}
$$

Due to rounding errors, your final answers for angles A and B may be slightly different to what is shown above. If your answers are within 0.5 degrees of the solutions given, you should consider your answer correct.

8. (a) We will start by finding the length of side c using the cosine rule.

$$c^2 = 17^2 + 9^2 - 2 \times 17 \times 9 \times \cos(118°)$$

Using our scientific calculator, we can evaluate the right-hand side of the above equation all in one step.

$$\begin{aligned} c^2 &= 513.66 \\ c &= \sqrt{513.66} \\ c &= 22.66\text{mm} \end{aligned}$$

We will now find the size of angle A using the sine rule.

$$\begin{aligned} \frac{\sin(A)}{17} &= \frac{\sin(118°)}{22.66} \\ \frac{\sin(A)}{17} \times 17 &= \frac{\sin(118°)}{22.66} \times 17 \\ \sin(A) &= \frac{\sin(118°)}{22.66} \times 17 \\ A &= \sin^{-1}\left[\frac{\sin(118°)}{22.66} \times 17\right] \\ A &= 41.48° \end{aligned}$$

Finally, we can find the size of angle B.

$$\begin{aligned} B &= 180° - 118° - 41.48° \\ &= 20.52° \end{aligned}$$

Due to rounding errors, your final answers for angles A and B may be slightly different to what is shown above.

(b) We will start by finding angle B using the sine rule.

$$\begin{aligned} \frac{\sin(B)}{16} &= \frac{\sin(56°)}{19} \\ \frac{\sin(B)}{16} \times 16 &= \frac{\sin(56°)}{19} \times 16 \\ \sin(B) &= \frac{\sin(56°)}{19} \times 16 \\ B &= \sin^{-1}\left[\frac{\sin(56°)}{19} \times 16\right] \\ B &= 44.28° \end{aligned}$$

We will now find the size of angle C.

$$\begin{aligned} B &= 180° - 56° - 44.28° \\ &= 79.72° \end{aligned}$$

Finally, we will find the length of side c using the sine rule.

$$\begin{aligned} \frac{c}{\sin(79.72°)} &= \frac{19}{\sin(56°)} \\ \frac{c}{\sin(79.72°)} \times \sin(79.72°) &= \frac{19}{\sin(56°)} \times \sin(79.72°) \\ c &= 22.55\text{cm} \end{aligned}$$

(c) As no angles are known, we will start by finding the size of angle C using the cosine rule.

$$\begin{aligned}
c^2 &= a^2 + b^2 - 2ab\cos(C) \\
28^2 &= 23^2 + 12^2 - 2 \times 23 \times 12 \times \cos(C) \\
784 &= 529 + 144 - 552\cos(C) \\
784 &= 673 - 552\cos(C) \\
784 - 673 &= 673 - 673 - 552\cos(C) \\
111 &= -552\cos(C) \\
\frac{111}{-552} &= \frac{-552\cos(C)}{-552} \\
-\frac{111}{552} &= \cos(C) \\
C &= \cos^{-1}\left(-\frac{111}{552}\right) \\
C &= 101.60^\circ
\end{aligned}$$

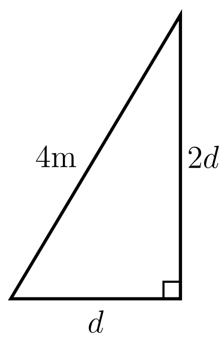
We will now find the size of angle A using the sine rule.

$$\begin{aligned}
\frac{\sin(A)}{23} &= \frac{\sin(101.60^\circ)}{28} \\
\frac{\sin(A)}{23} \times 23 &= \frac{\sin(101.60^\circ)}{28} \times 23 \\
\sin(A) &= \frac{\sin(101.60^\circ)}{28} \times 23 \\
A &= \sin^{-1}\left[\frac{\sin(101.60^\circ)}{28} \times 23\right] \\
A &= 53.58^\circ
\end{aligned}$$

Finally, we will find the size of angle B.

$$\begin{aligned}
B &= 180^\circ - 101.60^\circ - 53.58^\circ \\
&= 24.82^\circ
\end{aligned}$$

9. Let the distance between the base of the ladder and the wall be d.

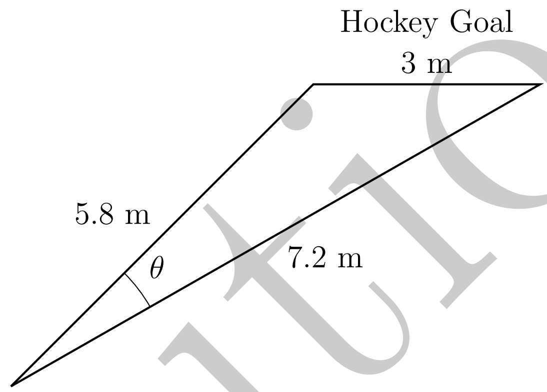


Using Pythagoras' Theorem, we find find the length of d.

$$\begin{aligned}
d^2 + (2d)^2 &= 16 \\
d^2 + 4d^2 &= 16 \\
5d^2 &= 16 \\
\frac{5d^2}{5} &= \frac{16}{5} \\
d^2 &= \frac{16}{5} \\
d &= \sqrt{\frac{16}{5}} \\
d &= 1.79 \text{ m}
\end{aligned}$$

Therefore, the base of the ladder is 1.79m from the wall.

10. We will start by drawing a diagram, including all known information. Note that the diagram is not to scale.



Using the cosine rule:

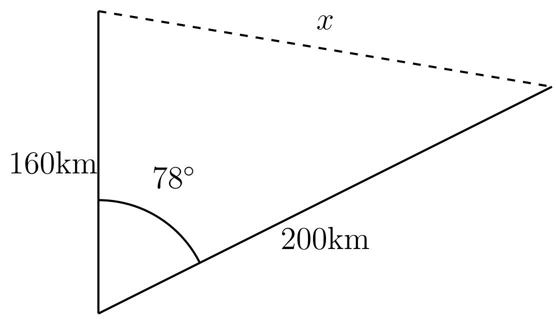
$$\begin{aligned}
3^2 &= 5.8^2 + 7.2^2 - 2 \times 5.8 \times 7.2 \times \cos(\theta) \\
9 &= 33.64 + 51.84 - 83.52\cos(\theta) \\
9 &= 85.48 - 83.52\cos(\theta) \\
9 - 85.48 &= 85.48 - 85.48 - 83.52\cos(\theta) \\
-76.48 &= -83.52\cos(\theta) \\
\frac{-76.48}{-83.52} &= \frac{-83.52\cos(\theta)}{-83.52} \\
\cos(\theta) &= \frac{76.48}{83.52} \\
\theta &= \cos^{-1}\left(\frac{76.48}{83.52}\right) \\
\theta &= 23.69^\circ
\end{aligned}$$

Therefore, the shot must be made within an angle of 23.69°.

11. The cars travelled for 2 hours.

$$\begin{aligned}
\text{Distance travelled by first car} &= 100 \times 2 = 200\text{km} \\
\text{Distance travelled by second car} &= 80 \times 2 = 160\text{km}
\end{aligned}$$

Let x be the distance between the two cars at 2pm.



We will now find the distance between the two cars, x, using the cosine rule.

$$x^2 = 160^2 + 200^2 - 2 \times 160 \times 200 \times \cos(78^\circ)$$

Using our scientific calculator, we can evaluate the right-hand side of the above equation all in one step.

$$x^2 = 52293.65$$
$$x = \sqrt{52293.65}$$
$$x = 228.68\text{km}$$

At 2pm, the cars are 228.68km apart.